SAMUEL ALEXANDER V

CYBERSECURITY SME | OFFENSIVE SECURITY & AI-DRIVEN SECURITY TOOLING

+91 75026 15599 | Uppillipalayam, Coimbatore | alexanderhere2005@gmail.com

ABOUT ME

Cybersecurity SME specializing in offensive security (web application penetration testing, AI-assisted red teaming, and secure software development). Independent security consultant with hands-on experience testing production environments for enterprise clients, uncovering critical-severity vulnerabilities including authentication bypass and injection flaws. Certified Ethical Hacker (CEH) with two patents and a published research draft on AI-driven SIEM automation. Ranked top 5% globally on TryHackMe. Builds custom offensive-security tooling — including an AI-orchestrated red-team automation framework — to accelerate assessment workflows. Available for freelance penetration testing, security audits, and secure development engagements.

WORK EXPERIENCE

Penetration Tester Intern | Montbleu, Coimbatore | 30 days

- Conducted black-box and grey-box web application penetration tests on live production environments.
- Identified and documented critical severity vulnerabilities including authentication bypass and injection flaws in production systems.
- Performed manual testing aligned with OWASP Top 10 methodology; delivered detailed PoC reports with remediation recommendations.
- Used Burp Suite Professional for traffic interception, active scanning, and manual exploitation (Intruder, Repeater, Collaborator).

AI Intern | Novitech, Coimbatore | 30 days

- Built face detection and object recognition pipelines using Python and OpenCV.
- Applied machine learning models for small-scale AI applications in a production-adjacent environment.

Embedded Systems & IoT Intern | Enthu Tech, Coimbatore | 14 days

- Developed IoT prototypes using microcontrollers; integrated sensor data pipelines.
- Contributed to AI-assisted face and object detection modules on embedded hardware.

TECHNICAL SKILLS

Security

Burp Suite , Metasploit, Wireshark, SQL Map, Hydra, John, Nmap, OWASP Top 10, MITRE ATT&CK, NVD, SIEM, ElasticSearch, AI-assisted red teaming

Programming

Python, Java, SQL, Javascript

Web / Backend

Django, Spring Boot, React, HTML/CSS, MySQL

DevOps / Tools

Git, Linux (Kali, Ubuntu), Docker, AWS, Google Cloud

PROJECTS

Gemini Red — AI-Orchestrated Red Team Automation Framework

CLI-based red-teaming orchestrator that executes offensive security tools in isolated containers on the host machine and streams structured output to an LLM (Gemini) for analysis and next-step decision-making.

Mr. Defender

NLP-based Kibana query generator integrated with SOAR for automated SOC task execution. Reduced manual alert triage time significantly.

Phishing URL Detection Extension
Cloud-based browser extension that classifies URLs as phishing or benign using CVSS and rule-based metrics in real time.

Secure E-Commerce Platform
Full-featured Django e-commerce site with security-first design: CSRF protection, parameterised queries, secure session management, input validation.

Face Detection using AI/ML
Python + OpenCV real-time face detection system demonstrating computer vision and ML skills.

EDUCATION

B.E. Computer Science & Engineering (Cyber Security)

Sri Krishna College of Engineering and Technology (SKCET), Coimbatore | 2023-2027 | CGPA: 8.2

Diploma in Computer Engineering

CIT Sandwich Polytechnic College, Coimbatore | 2021–2024 | 92%

RESEARCH & PATENTS

- **Patent**: An Intelligent Tamper-Evident Weighing Apparatus with Multi-Sensor Security and Data Authentication Framework
- **Patent**: Wearable Smart Air Filtration Helmet with AQI-Based Automatic Control and Filter Life Monitoring
- **Research Paper(Draft)**: Conversational SIEM that converts Natural Language to DSL queries

CERTIFICATIONS

- **Certified Ethical Hacker (CEH)** — CISCO
- **AWS Certified Developer – Associate** — Infosys Springboard
- **Google Cloud Computing Foundations** — Google
- **AI for App Building** — Google
- **Data Structures & Algorithms in Jav**a — Infosys Springboard

ACCOMPLISHMENTS

- **Top 4% globally —** TryHackMe (Cyber Security & Ethical Hacking Platform)
- **1st Place —** College-level Web Designing Competition, CIT Sandwich Polytechnic College
- **2nd Place —** Inter-college Paper Presentation on VR & AR, CIT Sandwich Polytechnic College

SOCIALS

- **LinkedIn —** https://www.linkedin.com/in/samuel-alexander-v-2b25712a5/
- **GitHub —** https://github.com/SamAlex-22
- **TryHackMe —** https://tryhackme.com/p/SamAlex